DENTAL CLINIC MANAGEMENT SYSTEM

REQUIREMENTS GATHERING AND ANALYSIS

System Analysis Document

Prepared for academic project use

Format: Black and White | Version: 1.0

This document describes the proposed requirements for a single-site dental clinic management system. Requirements should be validated with the clinic stakeholders before implementation.

Document Information

Item	Description
Document title	Dental Clinic Management System - Requirements Gathering and Analysis
Version	1.0
Document status	Initial requirements baseline for review
System type	Administrative and clinical information system for a dental clinic
Primary audience	Clinic owner, administrator, dentists, reception staff, cashier, project team
Purpose	Define the problem, scope, stakeholders, system behavior, data, quality needs, and acceptance conditions.

Contents

1. Executive Summary
2. Introduction and Background
3. Problem Statement
4. Project Objectives and Success Measures
5. Scope and Boundaries
6. Stakeholders and User Roles
7. Requirements Gathering Approach
8. Proposed Business Process
9. Functional Requirements
10. Use Cases and Scenarios
11. Non-Functional Requirements
12. Data Requirements and Data Dictionary
13. Business Rules
14. Access Control and Privacy
15. Reports and Outputs
16. Assumptions, Dependencies, and Constraints
17. Risks and Mitigation
18. Acceptance Criteria and Validation Plan
19. Future Enhancements
20. Glossary
21. Conclusion

1. Executive Summary

A Dental Clinic Management System (DCMS) is a software application intended to coordinate the routine administrative and clinical activities of a dental clinic. It centralizes patient registration, appointment scheduling, dentist availability, clinical notes, treatment history, invoices, payments, and operational reporting. Instead of relying on scattered paper files, notebooks, and separate spreadsheets, authorized staff use a consistent source of information.

The main need is not simply to digitize forms. The system should help staff find accurate information, reduce avoidable scheduling mistakes, preserve a history of care, make payment status clear, and support management decisions. It must also protect confidential patient information through login, role-based permissions, careful validation, and backup procedures.

This document provides a proposed requirements baseline for a first version designed for one clinic location. Features such as insurance claim processing, patient self-service, automated messaging, and integration with external medical systems are treated as possible future enhancements unless stakeholders explicitly include them in the initial release.

2. Introduction and Background

2.1 Purpose of the document

The purpose of this document is to record what the system is expected to do, who will use it, which data it will handle, and the conditions that will determine whether it is acceptable. It is intended to create a shared understanding between clinic stakeholders and the development team before detailed design, database construction, and programming begin.

2.2 Background

Dental clinics coordinate recurring appointments, clinical examinations, multiple treatment sessions, patient histories, and financial transactions. Information may be updated by receptionists, dentists, and cashiers at different times. If records are incomplete, duplicated, or difficult to retrieve, staff may spend unnecessary time searching, appointments may conflict, and outstanding balances may be misunderstood.

2.3 Definitions

- **Patient record:** demographic and contact information used to identify a patient.
- **Appointment:** a scheduled visit linked to a patient, a dentist, a date, and a time slot.
- **Clinical record:** notes about examination, diagnosis, procedure, treatment, and follow-up.
- **Invoice:** a financial record of charges for services provided or planned.
- **Requirement:** a documented capability, rule, or quality condition expected from the system.

3. Problem Statement

The clinic needs a reliable way to maintain related administrative, clinical, and financial information without depending on disconnected records. Common problems that the system is intended to address include:

- Patient details may be stored in separate paper files or spreadsheets, making updates and retrieval inconsistent.
- Manual scheduling can lead to overlapping appointments, incorrect time entries, or uncertainty about a dentist’s availability.
- Treatment details may be difficult to follow across multiple visits if notes are not stored in a consistent patient history.
- Billing records may not clearly show the services charged, payments received, or amount still due.
- Managers may lack timely summaries of daily appointments, new patients, completed treatments, or collected payments.
- Sensitive patient information can be exposed if files are accessible to people without a legitimate work-related need.

The proposed solution is a centralized system with clear user roles, structured records, validated workflows, and reports. The system should assist clinic staff rather than replace professional dental judgment; clinical decisions

remain the responsibility of qualified dental practitioners.

4. Project Objectives and Success Measures

Objective	Expected result / measure
Centralize patient information	Authorized staff can find an existing patient and view the current contact details without searching across separate files.
Improve appointment coordination	The system blocks conflicting bookings for the same dentist and time slot, except where an authorized, explicitly defined exception is allowed.
Maintain continuity of care	Dentists can review prior clinical notes and add a dated record for each relevant visit.
Improve financial visibility	Invoices, payments, and outstanding balances can be viewed and reconciled by authorized staff.
Improve accountability	Important records are associated with the user who created or updated them when audit tracking is enabled.
Support management	Users with permission can produce appointment and payment summaries for a selected period.
Protect confidentiality	Users must authenticate and access only the functions and records allowed by their role.

Success should be evaluated during user acceptance testing using realistic clinic scenarios. Numeric targets for response time, availability, recovery time, and record-retention periods should be agreed with the clinic because they depend on the deployment environment and applicable policies.

5. Scope and Boundaries

5.1 Included in the initial version

- Patient registration, profile editing, duplicate-check support, and patient search.
- Dentist profiles and working schedules, including availability and appointment status.
- Appointment booking, rescheduling, cancellation, check-in, and daily schedule viewing.
- Clinical visit records containing examination notes, diagnosis, procedures, treatment plan, and follow-up information.
- Service catalog and price reference, subject to configuration by authorized staff.
- Invoice creation, payment recording, balance calculation, and transaction history.
- User authentication and role-based access for administrators, receptionists, dentists, and cashiers.
- Basic operational reports and a backup/restore procedure appropriate to the selected technology.

5.2 Not included unless separately approved

- Automated diagnosis, AI-based treatment recommendations, or replacement of clinical judgment.
- Direct integration with insurance companies, laboratories, pharmacies, or external electronic health-record platforms.
- Online patient booking portal, mobile application, SMS/email gateway, or automated reminders.
- Online payment gateway or automatic bank reconciliation.
- Multi-branch management, inventory management for all dental supplies, payroll, and full accounting.
- Migration of historical paper records beyond the data and effort agreed for project delivery.

5.3 System boundary

The DCMS is responsible for storing and processing clinic records supplied by authorized users. Human users remain responsible for entering accurate information, confirming patient identity, recording clinical decisions, and verifying that financial entries reflect actual transactions. External services are outside the system boundary until an integration is explicitly specified and approved.

6. Stakeholders and User Roles

Stakeholder / role	Main needs	Typical permissions
Clinic owner / manager	Understand performance, configure clinic policies, review reports.	View management reports; approve settings and access rights.
System administrator	Maintain accounts, roles, configuration, and backups.	Manage users and configuration; technical access subject to policy.
Receptionist	Register patients, check schedules, book and update appointments.	Manage patient demographics and appointments; limited clinical/financial access.
Dentist	Review patient history, document examinations, diagnoses, and treatment.	View assigned patient information and create/update clinical notes according to policy.
Cashier / accountant	Create invoices, record payments, and inspect balances.	Manage billing and payment records; no need to edit clinical notes.
Patient	Provide accurate details and receive appointment information.	No direct system account in the initial version unless patient portal is approved.
Development / support team	Build, maintain, and troubleshoot the application.	Technical access should be controlled, logged where feasible, and limited to support needs.

A user may hold more than one role in a small clinic, but permissions should still be granted deliberately. Shared accounts should be avoided because they make it difficult to identify who performed an action.

7. Requirements Gathering Approach

Requirements should be confirmed through structured discussions with the people who perform the work, followed by review of the proposed workflows and example screens. The following techniques are recommended:

Technique	How it is used	Output
Stakeholder interviews	Ask the clinic owner, receptionist, dentists, and cashier about daily tasks, exceptions, and pain points.	Documented needs, priorities, and business rules.
Observation	Observe how appointments, visits, and payments are handled in practice, without unnecessarily exposing patient information.	Workflow steps, delays, and error-prone activities.
Document review	Review existing appointment books, anonymized forms, service lists, and invoice templates where available.	Required fields, terminology, and sample outputs.
Use-case discussion	Walk through normal and exceptional scenarios with representative users.	Actor goals, preconditions, steps, and alternate paths.
Prototype review	Present sample forms and schedules and ask users to identify missing or confusing fields.	Validated interface and workflow expectations.
Acceptance review	Test the agreed requirements with realistic test data before release.	Pass/fail outcomes and a list of issues to resolve.

All requirements should be assigned an identifier and a priority. A useful priority scheme is Must Have, Should Have, and Could Have. Any change after approval should be recorded with its reason, impact, owner, and approval status.

8. Proposed Business Process

8.1 Patient registration

- The receptionist searches for an existing patient using an identifier or suitable combination of name and contact details.
- If no matching patient is found, the receptionist creates a new patient record and enters required fields.
- The system validates required values and warns about likely duplicate records where practical.
- The receptionist confirms the record has been saved and can proceed to appointment booking.

8.2 Appointment lifecycle

- A staff member selects the patient, dentist, date, and available time slot.
- The system checks that the selected dentist is available and that the time slot does not conflict with an existing active appointment.
- The appointment is saved with an initial status, such as Scheduled.
- When the patient arrives, staff may mark the appointment as Checked In or In Progress according to clinic workflow.
- After the visit, the appointment can be marked Completed. If the patient cancels or does not attend, the status is updated rather than silently deleting the record.

8.3 Clinical visit

- The dentist confirms the patient identity and reviews relevant prior visits and medical alerts.
- The dentist creates a visit record containing the examination, diagnosis, procedure, treatment plan, and follow-up details as applicable.
- The system associates the record with the patient, dentist, and visit date, and preserves appropriate history.
- Any correction process should preserve record integrity and follow the clinic's approved policy.

8.4 Billing and payment

- Authorized staff select the relevant services or charges and create an invoice linked to the patient.
- The system calculates the total from the entered line items, discounts, and taxes only if these are configured and required.
- A cashier records the amount, date, and payment method for each payment received.
- The system calculates the remaining balance and shows whether the invoice is unpaid, partially paid, or paid.
- Financial corrections, voids, or refunds should be handled by an authorized user and leave an identifiable record.

9. Functional Requirements

Functional requirements describe observable services the system must provide. The following requirements are proposed for the first version and should be reviewed with stakeholders before being treated as final.

ID	Area	Requirement statement	Priority
FR-01	User authentication	The system shall require each staff user to sign in using an individual account before accessing protected functions.	Must Have
FR-02	Role management	The system shall support defined roles and allow only authorized administrators to create, disable, or change staff permissions.	Must Have
FR-03	Patient registration	The system shall allow authorized staff to create a patient record with a unique patient identifier and required demographic/contact fields.	Must Have
FR-04	Patient profile editing	The system shall allow authorized staff to update permitted patient details and retain the correct link between the patient and associated appointments, clinical records, and invoices.	Must Have
FR-05	Patient search	The system shall support searching by patient identifier and selected fields such as name or phone number, subject to access permissions.	Must Have
FR-06	Duplicate warning	The system should warn staff when entered details closely match an existing patient, while allowing authorized review rather than automatically merging records.	Should Have
FR-07	Dentist profiles	The system shall store dentist names, specialties where applicable, contact details, and active/inactive status.	Must Have
FR-08	Working schedule	Authorized users shall be able to define each dentist's working hours, unavailable periods, and exceptions.	Must Have
FR-09	Appointment booking	The system shall create an appointment linked to one patient and one dentist with a date, start time, duration, reason, and status.	Must Have

ID	Area	Requirement statement	Priority
FR-10	Conflict prevention	The system shall check dentist availability and prevent overlapping active appointments for the same dentist unless an explicitly approved exception rule applies.	Must Have
FR-11	Reschedule and cancel	The system shall allow authorized staff to change or cancel an appointment and record its current status.	Must Have
FR-12	Schedule views	The system shall display appointments by day and dentist, and allow users to filter by date, dentist, and status.	Must Have
FR-13	Visit status	The system should support practical statuses such as Scheduled, Checked In, In Progress, Completed, Cancelled, and No-Show.	Should Have
FR-14	Clinical visit note	The system shall allow an authorized dentist to create a dated visit note linked to the correct patient and dentist.	Must Have
FR-15	Treatment history	The system shall display the patient’s previous clinical records in chronological order to authorized dental staff.	Must Have
FR-16	Treatment plan	The system should allow the dentist to record proposed procedures, follow-up requirements, and relevant notes.	Should Have
FR-17	Service catalog	Authorized staff shall be able to maintain a list of clinic services and their current standard prices.	Should Have
FR-18	Invoice creation	The system shall create an invoice linked to a patient, with one or more charge line items, date, total amount, and status.	Must Have
FR-19	Payment recording	The system shall record payment amount, date, payment method, related invoice, and the staff account recording it.	Must Have
FR-20	Balance calculation	The system shall calculate the invoice balance from the total charges, applicable adjustments, and recorded payments.	Must Have
FR-21	Receipt / invoice view	The system should provide a printable invoice or receipt view with a unique reference number and a clear summary of charges and payments.	Should Have
FR-22	Operational reports	The system shall generate basic reports for appointments, patient registrations, invoices, payments, and outstanding balances over a selected period.	Must Have
FR-23	Input validation	The system shall validate mandatory fields, dates, numeric amounts, and relationships before saving records.	Must Have
FR-24	Audit information	The system should record creation and update timestamps and the responsible user for important records where supported by the design.	Should Have
FR-25	Backup and restore	The system shall support a documented way to back up application data and restore it after data loss or system failure.	Must Have
FR-26	Error messages	The system shall provide clear messages for invalid input, unavailable slots, access denial, and failed save operations.	Must Have
FR-27	Data export	The system could provide controlled export of selected reports to a common format for authorized users.	Could Have
FR-28	Appointment reminders	Automated SMS or email reminders may be considered after an approved messaging service and consent workflow are defined.	Could Have

10. Use Cases and Scenarios

10.1 Use Case UC-01: Register a patient

Field	Description
Primary actor	Receptionist
Goal	Create a reliable patient record.
Preconditions	The receptionist is signed in and has permission to manage patient demographics.
Main flow	1. Search for a possible existing record. 2. Review matching results. 3. Choose Add Patient if no suitable match exists. 4. Enter required information. 5. Save the record. 6. System validates the data and returns the patient identifier.
Alternative flows	If a likely duplicate is found, the system shows the match for review. If required data is missing, the system identifies the fields that need correction.
Postconditions	A new patient record is stored, or no record is created until validation succeeds.

10.2 Use Case UC-02: Book an appointment

Field	Description
Primary actor	Receptionist
Goal	Schedule a patient visit with an available dentist.
Preconditions	The patient exists and the dentist has a defined schedule.
Main flow	1. Select patient. 2. Select dentist and date. 3. Review available time slots. 4. Select a slot and enter visit reason/duration. 5. Confirm booking. 6. System checks availability again and saves the appointment.
Alternative flows	If the slot becomes unavailable, the system rejects the booking and asks the user to choose another slot. If the patient does not exist, the user must register or locate the patient first.
Postconditions	A scheduled appointment is saved without an unapproved conflict.

10.3 Use Case UC-03: Record a dental visit

Field	Description
Primary actor	Dentist
Goal	Record the clinical outcome of a patient visit.
Preconditions	The dentist is authenticated and authorized to access the patient record.
Main flow	1. Open the patient profile. 2. Review relevant history. 3. Select the current visit or create a visit record. 4. Enter examination findings, diagnosis, procedures, and follow-up notes as applicable. 5. Save the record. 6. System confirms successful storage.
Alternative flows	If a required field is missing, the system highlights it. If access is not permitted, the record remains protected.
Postconditions	A dated clinical record is associated with the patient and the responsible dentist.

10.4 Use Case UC-04: Record a payment

Field	Description
Primary actor	Cashier
Goal	Record money received and update the invoice balance.
Preconditions	The invoice exists and the cashier has billing permission.
Main flow	1. Find the invoice. 2. Review total and previous payments. 3. Enter payment amount, date, and method. 4. Confirm the transaction. 5. System records the payment and recalculates the balance. 6. Receipt view is made available if supported.
Alternative flows	The system rejects a negative amount or invalid input. If a payment exceeds the remaining balance, the system follows a clinic-approved rule rather than silently accepting an unexplained overpayment.
Postconditions	The payment is linked to the invoice and the updated balance/status is displayed.

11. Non-Functional Requirements

Non-functional requirements describe the quality and operational conditions of the system. The numeric values below are suggested starting points for stakeholder review and should be adjusted to the actual hosting environment, number of users, and clinic policy.

Category	Requirement / verification idea
Security	Protected features require authentication. Passwords must not be stored in plain text. Failed login attempts and account deactivation should be handled according to a defined policy.
Authorization	Each role must be tested against a permission matrix. A receptionist should not automatically gain permission to edit clinical notes or financial history.
Privacy	Patient details and clinical information must be visible only to staff with a legitimate role-based need. Data exports must be permission-controlled.
Performance	For the expected clinic workload, common searches and screen transitions should normally complete within an agreed target, suggested as 2-3 seconds on the local deployment.
Availability	The clinic should define expected operating hours and acceptable downtime. Planned maintenance should be communicated to staff.
Reliability	The system must not report a successful save unless the record has been persisted. Errors should not leave partial or contradictory records.
Data integrity	Appointments, visit notes, invoices, and payments must maintain valid references to existing records. Critical calculations should be validated.
Usability	Labels should be clear and consistent. Common tasks should require minimal navigation, and validation messages should explain how to fix errors.
Compatibility	The selected interface should work in the agreed browser/operating-system environment and at the screen sizes used by clinic staff.
Backup and recovery	Backups should be scheduled according to clinic risk tolerance, stored securely, and periodically tested by restoring a copy.
Maintainability	The design should separate patient, appointment, clinical, and financial modules so changes can be tested without unnecessary impact on other areas.
Auditability	For sensitive changes, the system should record who acted and when, where technically feasible. Audit data should also be protected from unauthorized alteration.
Accessibility	Forms should have readable contrast in normal use, understandable labels, and keyboard-accessible controls where supported by the interface.

12. Data Requirements and Data Dictionary

The following entities represent a proposed logical data model. Final field names and database types depend on the selected technology and the clinic’s approved workflow. Only necessary information should be collected, and access to sensitive fields must be controlled.

Entity	Suggested attributes	Purpose / relationships
Patient	PatientID, FullName, DateOfBirth, Gender (if required), Phone, Email (optional), Address (optional), EmergencyContact (if required), MedicalAlert, CreatedAt, IsActive	Identifies a patient. One patient may have many appointments, clinical records, and invoices.
Dentist	DentistID, FullName, Specialty, Phone, Email, IsActive	Identifies a dentist. One dentist may have many appointments and clinical records.
User	UserID, Username, PasswordHash, RoleID, IsActive, LastLoginAt	Represents a staff login. Credentials must be securely handled.
Role	RoleID, RoleName, PermissionSet or linked permissions	Defines access boundaries for a class of users.
DentistSchedule	ScheduleID, DentistID, Day/Date, StartTime, EndTime, AvailabilityStatus	Represents regular working hours, exceptions, or unavailable periods.
Appointment	AppointmentID, PatientID, DentistID, StartDateTime, Duration, Reason, Status, CreatedBy, CreatedAt, UpdatedAt	Links a patient and dentist to a visit slot.
ClinicalRecord	RecordID, PatientID, DentistID, AppointmentID (optional), VisitDate, ExaminationNotes, Diagnosis, Procedure, TreatmentPlan, FollowUp, CreatedAt, UpdatedAt	Stores the clinical notes for a visit.
Service	ServiceID, ServiceName, Description, StandardPrice, IsActive	Provides a configurable catalogue of clinic services.
Invoice	InvoiceID, PatientID, InvoiceDate, Subtotal, Discount (if used), Tax (if required), Total, Status, CreatedBy	Represents charges to a patient and may contain multiple invoice items.
InvoiceItem	InvoiceItemID, InvoiceID, ServiceID (optional), Description, Quantity, UnitPrice, LineTotal	Stores individual charges included in an invoice.
Payment	PaymentID, InvoiceID, Amount, PaymentDate, PaymentMethod, Reference (optional), RecordedBy, Status	Stores payments associated with an invoice.
AuditEvent (recommended)	AuditID, UserID, Action, EntityType, EntityID, Timestamp, Summary	Records relevant events for accountability where required and supported.

12.1 Data validation and integrity

- Each major record must have a unique identifier generated or checked by the system.
- Required patient, appointment, invoice, and payment fields must be validated before saving.
- Dates and times must be valid and internally consistent; an appointment end time must be later than its start time.
- An appointment must refer to an existing patient and dentist. An invoice item must refer to an existing invoice.
- Amounts must use an agreed currency and decimal precision. Negative payment values must not be accepted as ordinary payments.
- Historical clinical and financial records should not be silently deleted. Corrections should follow an approved amendment, void, or cancellation process.
- The system should define how inactive patients, dentists, services, and accounts are handled without breaking historical relationships.

13. Business Rules

ID	Rule
BR-01	Every appointment must be associated with one existing patient and one existing dentist.
BR-02	A dentist cannot have overlapping active appointments unless the clinic explicitly defines and approves an exception.
BR-03	Appointment duration must be positive and fit within the dentist's available working period, subject to approved exceptions.
BR-04	Cancelling an appointment changes its status; it does not erase the historical fact that the appointment was created.
BR-05	A clinical record must be associated with a patient and the responsible dentist.
BR-06	Only authorized dental staff may create or update clinical records.
BR-07	Each invoice must have a unique reference and one or more valid charge items, unless the clinic approves another billing workflow.
BR-08	The amount due equals the invoice total minus valid recorded payments and approved adjustments.
BR-09	A payment must be linked to an invoice and recorded with an amount, date, and method.
BR-10	Only designated roles may apply discounts, void invoices, reverse payments, or amend financial transactions.
BR-11	A user account that is disabled cannot sign in, but past actions associated with that account remain identifiable.
BR-12	A record cannot be permanently removed where doing so would break required financial or clinical history; any legally required deletion or retention process must be decided by the clinic.

14. Access Control and Privacy

The system will contain personal and potentially sensitive health information. Security and privacy rules must therefore be considered from the beginning of the design, not added as an afterthought. The exact legal obligations, consent requirements, and retention periods should be confirmed with the clinic and qualified local compliance guidance before deployment.

14.1 Initial role-permission matrix

Function	Admin	Receptionist	Dentist	Cashier
Manage users and roles	Yes	No	No	No
View/edit patient demographics	Yes	Yes	View / limited edit	View as needed
Manage appointments	Yes	Yes	View own schedule / update visit status as approved	No
Read/write clinical records	Controlled admin access	No by default	Yes, as authorized	No
Create invoices	Yes / approved	No or limited	View relevant details	Yes
Record payments	Yes / approved	No or limited	No	Yes
View financial reports	Yes	No or limited	Only if required	Yes, within role
Configure backup/system settings	Yes / designated admin	No	No	No

This matrix is an initial proposal. A clinic may combine duties in practice, but it should explicitly approve the resulting permissions. Administrative access to clinical information should be limited to what is needed for support and management.

14.2 Minimum security expectations

- Use individual accounts and secure password storage; never store or display plain-text passwords.
- Enforce role-based access on the server or application logic, not only by hiding buttons on the screen.

- Use secure connections when data travels across a network, where the deployment supports them.
- Restrict backups and exported files to authorized staff and protect them from loss or unauthorized access.
- Provide a process to disable accounts quickly when a staff member leaves or changes responsibilities.
- Avoid displaying more patient information than necessary on shared screens, printed reports, or exports.

15. Reports and Outputs

Report / output	Main fields or measures	Intended users
Daily appointment schedule	Appointment time, patient, dentist, visit reason, and status.	Receptionist, dentist, manager
Patient summary	Patient identity and contact information, relevant alerts, appointment list, and clinical-history references.	Authorized clinical staff
Appointment activity report	Counts of scheduled, completed, cancelled, and no-show appointments over a date range.	Manager, receptionist
Invoice statement	Invoice number, services/charges, total, payments, balance, and status.	Cashier, authorized manager, patient when appropriate
Payment summary	Payment date, amount, method, related invoice, and recording user.	Cashier, manager
Outstanding balances	Invoices with an unpaid or partially paid balance and their age/status.	Cashier, manager
User activity summary (if audit feature is implemented)	Selected important actions, user, timestamp, and affected record.	Authorized administrator / manager

Reports should show the period covered, the date they were generated, and any applied filters. Sensitive reports should be limited to authorized users and should not expose clinical notes when only financial totals are required.

16. Assumptions, Dependencies, and Constraints

16.1 Assumptions

- The first release is intended for one clinic location.
- Clinic staff will receive basic training and will be responsible for entering accurate information.
- The clinic will decide appointment durations, working hours, service prices, currency, and permitted payment methods.
- The clinic will define which demographic and clinical fields are mandatory and who may edit them.
- The computer/network environment will meet the requirements of the selected application and database.

16.2 Dependencies

- A database and hosting or local-server environment must be selected and maintained.
- Secure authentication and backup mechanisms depend on the chosen implementation and operating environment.
- External reminders, online payments, or insurance workflows require approved third-party services and separate integration requirements.
- Historical data import depends on the availability, quality, format, and permission to use existing records.

16.3 Constraints

- Project time, budget, team size, and available hardware may limit the first-release feature set.
- Data migration and user training may require clinic staff time outside normal patient-care activities.
- Privacy, retention, and financial handling rules must be confirmed before real patient data is loaded.

• The system must not be presented as making clinical decisions or guaranteeing medical outcomes.

17. Risks and Mitigation

Risk	Possible impact	Mitigation
Incorrect or incomplete data	Wrong patient information, scheduling mistakes, or unreliable reports.	Validate required fields, train users, provide duplicate warnings, and review data quality.
Unauthorized access	Exposure or alteration of patient, clinical, or financial information.	Individual accounts, role-based permissions, secure authentication, account disabling, and audit records.
Database loss or corruption	Disruption to clinic operation and potential loss of important records.	Automated backups, restricted backup access, restore testing, and documented recovery responsibilities.
Schedule conflicts	Patient inconvenience and lost clinic time.	Recheck availability when saving, define appointment durations, and manage schedule exceptions explicitly.
Incorrect payments or balances	Disputes, inaccurate income summaries, or reconciliation effort.	Validate amounts, link payments to invoices, restrict financial adjustments, and retain transaction history.
Low staff adoption	Users may continue using separate records, creating inconsistent data.	Involve staff in requirements review, make workflows practical, provide training, and collect feedback.
Unclear project scope	Delays or incomplete delivery due to expanding requirements.	Approve an initial scope, prioritize requirements, and use a change-request process.

18. Acceptance Criteria and Validation Plan

A requirement is accepted only when the corresponding behavior can be demonstrated and verified. Testing should use fictional or properly anonymized test data, not live patient records unless the clinic has approved a controlled test process.

Test ID	Scenario	Expected result
AT-01	Create a patient with all required fields.	The system saves the record and returns a unique patient identifier.
AT-02	Attempt to save a patient while a required field is missing.	The system prevents an invalid save and identifies the missing information.
AT-03	Search for a patient by identifier or supported field.	The appropriate patient record is returned only to an authorized user.
AT-04	Book an available time slot.	The appointment is saved and shown on the correct schedule.
AT-05	Try to book an overlapping active appointment for the same dentist.	The system blocks the conflict or requires a specifically approved exception.
AT-06	Reschedule or cancel an existing appointment.	The updated status/time is stored and the change is reflected in schedule views.
AT-07	Create a clinical record as an authorized dentist.	The saved record is linked to the correct patient and dentist and appears in history.
AT-08	Attempt to edit clinical notes using a role without permission.	The system denies the action and does not alter the record.
AT-09	Create an invoice with multiple items.	The invoice total is calculated according to configured rules and stored with a unique reference.
AT-10	Record a valid partial payment.	The payment is saved and the remaining balance/status are recalculated correctly.
AT-11	Attempt to enter a negative payment or invalid amount.	The system rejects the input or follows a separately approved adjustment workflow.
AT-12	Generate an appointment or payment report for a date range.	The report applies the requested range and displays consistent totals and filters.
AT-13	Disable a staff account and attempt sign-in.	The disabled account cannot access the system.
AT-14	Perform a backup and restore test in a controlled environment.	The restored data is usable and the recovery procedure is documented.

18.1 Approval checklist

- Clinic representatives have reviewed the scope and the user roles.
- Required data fields and sensitive information rules have been agreed.
- Appointment-conflict handling and cancellation statuses have been confirmed.
- Billing rules, discounts, refunds, and payment methods have been confirmed.
- Security, backup, and reporting expectations have named owners.
- Open questions and deferred features are recorded before development is finalized.

19. Future Enhancements

The following ideas may provide value after the core workflow is stable. Each feature should be separately assessed for cost, security, privacy, reliability, and user consent before being added:

- Patient portal for requesting or confirming appointments and viewing approved information.
- SMS or email reminders with configurable timing and opt-in/consent handling.
- Inventory tracking for dental materials, instruments, and consumables.
- Insurance information and claim tracking, subject to the relevant payer processes.
- Online payment integration and automated reconciliation.
- Multi-branch support with controlled access across clinic locations.
- Analytics dashboards for appointment utilization, revenue trends, and service demand.
- Integration with approved imaging systems or external electronic health-record tools.

20. Glossary

Term	Meaning
Authentication	Verifying a user’s identity before access is granted.
Authorization	Determining which functions and data an authenticated user may access.
Clinical history	The sequence of recorded examinations, diagnoses, treatments, and follow-up information for a patient.
Functional requirement	A capability or action the system must provide.
Non-functional requirement	A quality, security, performance, or operational condition for the system.
Role-based access control	A permission model in which users receive access according to assigned roles.
Acceptance criteria	Specific conditions used to decide whether a requirement has been implemented correctly.
Audit trail	A protected history of selected actions, including who performed an action and when.
Data integrity	The accuracy, consistency, and validity of data and its relationships.

21. Conclusion

This expanded requirements document establishes a structured starting point for the Dental Clinic Management System. It describes the business problem, project boundaries, stakeholders, gathering approach, workflows, detailed functional requirements, use cases, quality expectations, data entities, business rules, privacy controls, reports, risks, and acceptance tests.

Before implementation, clinic representatives should approve the requirements and resolve key policies for clinical access, data retention, financial adjustments, and backup recovery. The approved document can then guide design, implementation, testing, and acceptance.